■ RetailPulse

AI-Powered Customer Analytics & Demand Forecasting Platform

Advanced Data Science & Analytics Project · Zidio Development

Predictive Demand · Customer Segmentation · Churn Analysis · Inventory Optimization

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Version	2.0 — Industry Edition
Dataset	UCI Online Retail II
Stack	Python · pandas · scikit-learn · Streamlit · MLflow

1. Project Overview

RetailPulse is an end-to-end analytics platform that ingests raw retail transactions, cleans them with explainable policy decisions, builds defensible RFM segments, produces 30-day demand forecasts evaluated against seasonal-naive baselines, predicts customer churn risk with no temporal leakage, and feeds the demand signal into a configurable inventory recommendation engine. Every dashboard page is rendered from real computed parquet artefacts — no hardcoded KPI placeholders.

Quantified impact targets

Outcome	Target	This build's measured value
Forecast model vs baseline	Beat naive	30.9 % WAPE vs 69.2 % (aggregate)
Churn AUC	≥ 0.88 (goal)	AUC-ROC = 0.826 (RF), 0.817 (logistic)
Customer segments	6–8	8 RFM segments
Forecast horizon	≥ 30 days	30 days
Pipeline run time	≤ 5 min	~85 seconds end-to-end

Non-functional

Requirement	Status
CPU-only laptop compatible	Yes — scikit-learn, no GPU
Reproducible retraining	Yes — `python -m src.build_processed`
Experiment tracking	Yes — MLflow local file store
Drift monitoring	Yes — PSI per RFM feature
Streamlit dashboard	Yes — 6 pages, sidebar nav

2. Dataset & Data Dictionary

Source workbook: **online_retail_II.xlsx** (UCI Online Retail II). Two annual sheets — *Year 2009-2010* and *Year 2010-2011* — read in streaming mode with openpyxl to avoid loading the entire 45 MB workbook into RAM at once.

Raw observations

Metric	Value
Raw rows (both sheets)	1,067,371
Cancellations (Invoice starts with C)	19,433
Cleaned sales rows	793,609
Unique customers	5,878
Unique products	4,631
Unique invoices	36,969
Total gross revenue (GB£)	£17,685,460.64
Date range	2009-12-01 → 2011-12-09

Data dictionary

Column	Type	Description
Invoice	string	Invoice number. 'C' prefix → cancellation.
StockCode	string	Product code. 5 digits or alpha-numeric (e.g. M, POST).
Description	string	Product name.
Quantity	int	Units per line. Negative ⇒ cancellation/return.
InvoiceDate	datetime	Transaction timestamp.
Price	float (£)	Unit price (GBP).
Customer ID	float	Customer identifier; nullable.
Country	string	Billing country.

3. Data Cleaning & EDA

Policy decisions documented in *src/data_cleaning.py*: cancellations are **flagged not deleted**, negative quantities on non-cancelled invoices are separated, rows with missing Customer ID on non-cancelled invoices are kept for revenue analysis but excluded from customer-level RFM and churn.

Cleaning summary

Transformation	Decision
Duplicates	Drop (deterministic)
Cancellation invoices (C-prefix)	Retain separately, exclude from gross revenue
Negative Quantity on non-cancelled	Exclude from clean sales (returns path)
Missing Customer ID + non-cancel	Exclude from RFM / churn, retain in sales
Date parsing	ISO 8601, coerce errors → NaN
Numeric coercion	Quantity, Price → numeric; errors → NaN

EDA highlights

Britain (the United Kingdom) is the dominant market — revenue concentration approaches 80 % of the total. Monthly revenue shows clear seasonality: December spikes, February trough. Top products (e.g. WHITE HANGING HEART T-LIGHT HOLDER, POPCORN HOLDER, REGENCY CAKESTAND 3 TIER, JUMBO BAG RED RETROSPOT) collectively account for a meaningful share of gross revenue. The cancellation rate sits around 1.8 % of all invoices.

4. Customer Segmentation (RFM + K-Means)

We score customers on Recency, Frequency and Monetary value using quintile qcut (1–5). The Recency score is reverse-ordered so freshness gets a high score. K-Means on the standardised (Recency, Frequency, log1p-Monetary) feature space picks the best k via silhouette; DBSCAN is also computed for comparison.

Measured RFM result

Result	Value
Customers in RFM table	5,878
RFM business segments	8 (Champions · Loyal · Recent · At-Risk VIP · At-Risk · Low-Value Recent · Hibernating · Potential Loyal
K-Means best k	5
K-Means silhouette	0.443

Segment distribution

Segment	Customers
Hibernating	1527
Champions	1291
Potential Loyalists	1117
Loyal Customers	607
Recent Customers	441
At-Risk Customers	389
Low-Value Recent	280
At-Risk VIPs	226

5. Demand Forecasting

Two-stage approach: a seasonal-naive baseline (7-day repeat) and a HistGradientBoostingRegressor on calendar + lag + rolling features. Metrics are computed on a chronological hold-out of the last 30 days — no random shuffle, no future leakage. Prophet is installed and tried as a reference; on this dataset it did not improve on the gradient booster for the 30-day horizon.

Aggregate 30-day hold-out (chronological)

Metric	Seasonal-naive	HistGradientBoosting	Ensemble (avg)
MAE	19,433	8,681	12,274
RMSE	28,712	13,962	18,092
WAPE	69.2%	30.9%	43.7%

LSTM was deliberately excluded from the production path: per the master prompt rule and CPU-only constraints, tabular gradient boosters and well-engineered calendar features routinely outperform RNNs on this kind of sparse-dispatch retail data.

Per-product 30-day hold-out (top 10 by revenue)

StockCode	Naive MAE	Model MAE	Naive WAPE	Model WAPE
22423	57.2	54.5	150.7%	143.6%
85123A	124.3	144.2	88.9%	103.1%
85099B	198.0	190.9	100.5%	96.9%
M	264.7	192.4	141.0%	102.5%
84879	150.6	178.1	79.0%	93.5%
POST	8.3	7.0	44.0%	36.9%
47566	25.4	26.8	84.6%	89.0%
23166	43.3	30.0	147.6%	102.3%
22086	115.2	107.4	59.5%	55.5%

6. Customer Churn

The dataset has no explicit churn label. We construct a time-aware churn definition: split at a chronological cutoff, observe customer behaviour up to the cutoff, and label a customer **churned** if they have no purchase within the future window and their last observed purchase is more than a threshold beyond the cutoff. Features never use the future window.

Time-aware labelling summary

Parameter	Value
Observation cutoff	2011-07-28
Future prediction window (days)	90
Churn threshold (days beyond cutoff)	60
Customers observed	5,126
Labelled churned	3,212
Churn rate	62.66%

Model evaluation

Model	AUC-ROC	F1	Precision	Recall	Prec@Top20%
Logistic Regression	0.817	0.786	0.836	0.741	—
Random Forest	0.826	0.809	0.813	0.806	0.922

RandomForest beats the logistic baseline on AUC (≈ +0.009) and adds Precision@Top20 % ≈ 0.92, giving prioritised retention outreach a strong precision lift over the customer base.

7. Inventory Optimization

The Online Retail II dataset does NOT include on-hand stock, supplier lead time or purchase orders. The inventory recommendation is therefore an **estimate-based** plan that combines the Page-3 forecast with user-supplied on-hand, lead time and service level. The reorder math is documented and reproducible.

Replenishment formula

ReorderPoint (ROP) = $E[D_L] + z \cdot \sigma_D \cdot \sqrt{L}$
RecommendedReorderQty = max(0, ROP – OnHand)

- $E[D_L]$ = daily forecast · lead time L
- σ_D = historical daily standard deviation
- z = z-score from chosen service level (0.80–0.999)

Inventory plan summary (default: 7-day lead, 95 % SL)

Metric	Value
Products planned	4,631
Stockout risk flags	4497
Total recommended units	1,017,492
Median ROP	97.45

8. Streamlit Dashboard

Six pages with sidebar navigation. Each page renders KPIs and Plotly charts from the parquet artefacts we computed in src/build_processed.py. The pages can also be exported as static HTML (see docs/01_sales.html … docs/06_monitoring.html).

Dashboard pages

#	Page	Key views
1	Sales	Total revenue, invoices, AOV, monthly trend, top products, country mix
2	Customer Analytics	8 RFM segments, K-Means silhouette, segment contribution
3	Demand Forecasting	30-day aggregate + per-product forecast, baseline vs model
4	Churn Risk	Risk-band histogram, top-150 customers, retention guidance
5	Inventory	Configurable lead/SL/on-hand, reorder math, stockout flags
6	MLOps	PSI drift per feature, MLflow runs, retraining instructions

9. MLOps & Drift Monitoring

Lightweight, transparent drift monitor using the Population Stability Index (PSI) on log-scaled RFM features, with reference = earliest 50 % of customers and current = latest 50 %. PSI < 0.10 ⇒ stable, 0.10–0.25 ⇒ moderate drift, > 0.25 ⇒ high drift.

Drift report

Feature	PSI	Status
RecencyDays	12.4578	high drift
Frequency	0.9048	high drift
Monetary	0.825	high drift

MLflow

Three experiments logged locally to *mlruns/*: rfm_kmeans, forecast_aggregate and churn. Each logs parameters and metrics; the storage URI is *file:./mlruns* with *MLFLOW_ALLOW_FILE_STORE=true* set during this build.

10. Testing

pytest suite in *tests/test_retailpulse.py* covers cleaning, RFM scoring, seasonal-naive shape, WAPE/MAPE, PSI behaviour and inventory formula columns.

Test	Validates
test_basic_clean_segments	Cancellation flagging & clean sales subset
test_rfm_pipeline	End-to-end RFM table → scores → segments
test_seasonal_naive_repeats	Baseline forecast length & wrap-around
test_wape_zero_when_perfect	WAPE = 0 on identical series
test_psi_stable_when_identical	PSI near zero on identical distributions
test_psi_drift_when_distributions_shift	PSI increases when target shifts
test_recommend_returns_columns	Inventory returns required columns

11. Limitations & Honest Disclosures

• AUC-ROC on the chronological hold-out reached 0.826 — below the aspirational 0.88 target. We do not inflate the metric with random shuffles or target leakage.

• MAPE was not under 12 % on the aggregate (≈ 31 % on this very spiky daily dispatch-volume series). Per-product WAPE is delivered for products with enough history; for products without, we explicitly mark 'insufficient history' rather than fabricate.

• Online Retail II does not contain stock on hand, supplier lead times or purchase orders. Inventory recommendations depend on user-supplied on-hand, lead time and service level — clearly labelled in the dashboard.

• LSTM was kept out of the production path per master-prompt rule (CPU-only, sparse-dispatch dataset). Prophet is installed and tried but did not improve on the gradient booster at the 30-day horizon.

12. Personal Reflection (growth)

Building this project taught me to choose simple models that hold up under rigorous evaluation rather than chasing headline metrics. The biggest wins were in data discipline (cancellation policy, leak-free churn labels) and in translating model outputs into business decisions (RFM business segments, inventory reorder math). Next steps would be richer drift detection (Evidently AI), Optuna-driven hyper-parameter tuning and GCP/Fly.io deployment with a CI test-and-build pipeline.

13. Deliverables Summary

Deliverable	Status
Project report (A4 PDF, 10–18 pages)	Included — `docs/RetailPulse_Report.pdf`
GitHub repository	Code is on disk at /home/user/RetailPulse, ready to push
README.md	Included at repo root
Live public demo URL	Not deployed from this sandbox; Dockerfile + Streamlit config provided
Demo video (4–8 min)	Script in README; recording on a real machine